

March 2021

THERMAL ENGINEERING – I

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and Answer any five questions from the rest.

1. Answer the following questions (any twenty):

1×20

State True or False:

- i) During throttling process enthalpy remains constant.
- ii) H-S diagram for water steam is called Mollier diagram.
- iii) Superheated steam behaves like a perfect gas.
- iv) During heat transfer by radiation no medium is required.
- v) Intensive properties per unit mass are extensive properties.
- vi) A process with identical end states is called a cycle.
- vii) PMM-II violates the first law of thermodynamics.

Choose correct answer from the given alternatives:

- viii) Which of the following relation is correct? -
a) $C_p/C_v=R$ b) $C_p+C_v=R$ c) $C_p/C_v=\gamma$ d) $C_p-C_v=\gamma$
- ix) For same temperature limit which relation will be correct? -
a) $1 - \text{COP} = \eta_{\text{HE}}$ b) $1 + \text{COP} = \eta_{\text{HE}}$
c) $\text{COP} - \eta_{\text{HE}} = 1$ d) None of these
- x) The work transfer for isochoric process will be -
a) $P_1 V_1 \ln P_1/P_2$ b) $(P_1 V_1 - P_2 V_2)/(N-1)$
c) Zero d) None of these
- xi) For which of the following processes Heat Transfer will be zero? -
a) Adiabatic b) Isochoric c) Isobaric d) None of these
- xii) Dryness fraction of wet saturated steam will be -
a) >1 b) <1 c) $=1$ d) 0
- xiii) Which of the following relation is true? -
a) $1 = \alpha + \beta + \gamma$ b) $\alpha + \beta - 2\gamma = 2$ c) $\alpha - \beta - \gamma = 0$ d) $\alpha\beta = \gamma$
- xiv) Which of the following is renewable source of energy? -
a) Water b) Coal c) Natural Gas d) None of these
- xv) Turbine are used in following power plant -
a) Solar b) Thermal c) Nuclear d) Both b & c
- xvi) A cup of tea place on a table is an example of -
a) Open system b) Closed System
c) Isolated system d) None of these
- xvii) Which will be is unit of Power given below? -
a) KJ/Kg b) KJ/hr c) Cal/Kg d) KJ-hr

Fill in the blanks:

- xviii) The value of universal gas constant is _____.
xix) The work done in free expansion process is _____.
xx) The locus of saturated liquid line and saturated vapor line meets at _____.
xxi) The latent heat of steam at atmospheric pressure is _____.
xxii) The thermal conductivity of solid metals _____ with rise in temperature.
xxiii) Summation of internal energy and _____ is called enthalpy.
xxiv) Atmospheric pressure is equal to _____ bar.
xxv) Kelvin-Planks law deals with _____.
2. Define heat engine, Refrigerator, Heat pump. Prove Reversible engines working between two fixed temperature always be more efficient than any other engine working between same two fixed temperatures. 6+4
3. 0.45 kg of Air at 179°C expands adiabatically and reversibly to three times its original volume and during the process there is a fall in temperature to 15.5 °C. The work done during the process is 53 kJ. Calculate C_p and C_v . Derive the expression of Enthalpy for constant pressure non flow process. 7+3
4. Define Dryness fraction, Triple point, Critical point, Super heated steam. Discuss phase transformation with diagram. (4x2)+2
5. Find the enthalpy, Internal Energy, and Entropy of 1 kg of steam at a pressure of 10 bar when steam is (a) Dry saturated, (b) When steam is .75 Dry, (c) When steam is at 250°C .Use steam table. 10
6. Classify energy sources. Discuss disadvantages of Renewable energy. Write short notes on Geothermal Energy, Fossil Fuel, Green Energy. 2+2+6
7. A non flow Quasi static process (Reversible) occurs for which $p = (-3V + 16)$ bar, where V is volume in m^3 . What is the work done if V changes from 2 to 6 m^3 . Derive $PV^x = \text{Const.}$ (with usual notation) 4+6
8. Discuss Fourier's Law of Conduction. Derive an expression of conduction through a flat composite wall of several homogeneous layers. 3+7
9. Discuss cross flow heat exchanger. Define Grey body, Black body, Absorptivity, Stefan Boltzman law. 2+(4x2)
10. Discuss Zeroth law of Thermodynamics. Derive Steady flow energy equation. Write down short notes on Intensive and Extensive properties, quasi static process. 2+4+4
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